

Problem Statement and Goals

CADENCE

Team #4, CADENCE

Anam Khan
Haniya Kashif
Vivian Nguyen
Kelsey Rough
Hareem Salman

Table 1: Revision History

Date	Developer(s)	Change
Date1	Name(s)	Description of changes
Date2	Name(s)	Description of changes
...	...	...

1 Problem Statement

[You should check your problem statement with the [problem statement checklist](#) —SS].

[You can change the section headings, as long as you include the required information. —SS]

1.1 Problem

[What problem are you trying to solve? Why is it important? Do not talk about how you will solve the problem, only what the problem is. AI is rarely part of the problem. The problem is “what” you want to do, not “how” you want to do it. The problem statement should be written in a way that is understandable to a non-technical audience. —SS]

We want to provide a system that can accept the Doppler output from an ultrasound machine, determine a measure of mean blood velocity from that signal, and provide a continuous output suitable for recording and analysis. The system should produce measurements that are sufficiently accurate and reliable for laboratory research use.

1.2 Inputs and Outputs

The primary input to the system is the Doppler ultrasound audio signal representing blood flow within an artery.

The primary output is a continuous signal representing mean blood velocity that can be recorded and analyzed using LabChart. The system should also provide sufficient information to allow researchers to verify that the resulting measurements are reasonable and reliable.

1.3 Stakeholders

The primary stakeholders are the researchers and laboratory staff who require MBV measurements for physiological research. The project supervisors and biomedical researchers are also stakeholders, as they will provide domain knowledge, define research requirements, and evaluate the system's results. Ultrasound operators are stakeholders because the system must work with the output produced by the laboratory's ultrasound equipment. Future research participants are indirect stakeholders because the system may eventually be used to obtain measurements during human research studies.

1.4 Environment

[Hardware and Software Environment for the users. The developer environment is summarized as part of the developer plan. —SS]

The system will operate in a laboratory environment alongside an existing Doppler ultrasound system and LabChart data acquisition software. Researchers will use the system to obtain continuous MBV measurements during physiological experiments. The system will require appropriate hardware for receiving the ultrasound audio output and producing a signal that can be recorded by LabChart.

2 Goals

The primary goal of this project is to develop and validate a functional digital signal processing system that can derive mean blood velocity from the Doppler ultrasound audio output and provide a continuous signal compatible with LabChart.

The system goals are to:

1. Receive and process the audio output from a Doppler ultrasound system in real time.
2. Investigate and implement appropriate signal-processing techniques for extracting information related to blood velocity.
3. Calculate a continuous estimate of mean blood velocity from the processed Doppler signal.

4. Generate an output signal that can be recorded and analyzed using LabChart.
5. Formally validating the system's output against a ground truth, appropriate reference measurements or established methods.
6. Provide documentation describing the system, its operation, and its limitations.

3 Stretch Goals

[Goals you hope to get to, but not necessary. They are also helpful if the scope of the original project needs to be expanded. —SS]

Additional goals we'd hope to implement in CADENCE include but are not limited to the following: 1. Using the same input signal from the ultrasound, we can apply different signal transformation algorithms to derive metrics other than just MBV, such as Peak systolic velocity (PSV), End-diastolic velocity (EDV), Pulsatility Index (PI), and Resistance Index (RI) to gain further insight into the research participant's cardiovascular health. 2. Alongside the LabChart output, the system displays a real-time spectrogram (a color-coded time-frequency display) to provide a more insightful diagnostic trail for research purposes.

4 Custom Documents (CDs)

[For CAS 741: State whether the project is a research project. This designation, with the approval (or request) of the instructor, can be modified over the course of the term. —SS]

[For SE Capstone: List your custom documents (CDs). Potential CDs include usability testing, code walkthroughs, user documentation, formal proof, GenderMag personas, Design Thinking, etc. (The full list is on the course outline and in Lecture 02.) Normally the number of CDs will be two (one per term). Approval of the CDs will be part of the discussion with the instructor for approving the project. The CDs, with the approval (or request) of the instructor, can be modified over the course of the term. —SS]

Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?